

MEGA-GAS TRANSMISSION SYSTEM (120 BAR) - 56" X80M PSL 2 QUALITY PLAN (ITP)

Pipe & Coating Description: 56" (1422 mm) x 25.40 mm, Grade X80M PSL 2, SAWL / SAWH (Heavy-Wall Ultra-High Pressure Mainline) | **Governing Codes:** API Spec 5L 47th Ed. / ISO 3183 PSL 2 / High-Pressure Gas Pipeline Code

No	Inspection / Test Stage	Frequency & Sampling	Location & Specimen	Test Standard & Procedure	Specified Acceptance Criteria & Tolerances	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis (Ultra-Clean)	One analysis per heat of steel	Ladle vacuum sample	ASTM A751 / ISO 14284	C <= 0.07%, Si <= 0.35%, Mn: 1.65-1.90%, P <= 0.012%, S <= 0.0020%, Nb+V+Ti <= 0.15%, CE_Pcm <= 0.19, CE_IIW <= 0.40	API 5L Tablo 5	H/W
2	Product Chemical Analysis	Two analyses per test unit (lot <= 50 pipes)	Pipe body coupon	ISO 14284 / OES	C <= 0.07%, P <= 0.012%, S <= 0.0020%, CE_Pcm <= 0.19 (verifying strict low-carbon weldability)	API 5L Tablo 5	W
3	Pipe Body Heavy-Wall Tensile Test	Once per test unit (lot of <= 50 pipes)	Transverse rectangular strap	ISO 6892-1 / ASTM A370	Rt0.5: 555 - 705 MPa (80.5-102.2 ksi), Rm: 625 - 825 MPa, Af >= 18.0%, Yield-to-Tensile Ratio Y/T <= 0.90	API 5L Tablo 7	H/W
4	Weld Seam Tensile Test	Once per test unit	Transverse weld coupon	ASTM A370	Tensile strength Rm >= 625 MPa across submerged arc weld seam (failure must occur outside weld)	API 5L Tablo 7	W
5	Ultra-Tough CVN Charpy Impact Test at -20 °C	1 set body + 1 set weld + 1 set HAZ at -20 °C	Transverse Charpy V (10x10 mm)	ISO 148-1 / ASTM A370	At -20 °C: Body Min Avg 100 Joules, Min Ind 80 Joules; Weld & HAZ: Min Avg 60 Joules, Min Ind 45 Joules	API 5L Tablo 8 & Spec	H/W
6	Drop Weight Tear Test (DWTT) at -10 °C	Once per heat at -10 °C (2 specimens)	Transverse full thickness DWTT	API RP 5L3 / ASTM E436	At -10 °C: Average shear fracture area >= 85%, No single specimen < 70% (arresting brittle running fractures)	API 5L Tablo 18	H/W
7	Guided-Bend Test (Side Bend for Heavy-Wall)	2 side-bend specimens per test unit	Transverse weld cross section	ISO 5173 / ASTM A370	180° bend around mandrel; ZERO crack opening > 3.2 mm in weld metal or fusion boundary HAZ	API 5L 9.10	W
8	Macro Hardness Survey (HV10)	Once per test unit (16-point survey)	Cross section macro etch	ISO 6507-1 / ASTM E384	Maximum 280 HV10 across Body, HAZ, Inner weld and Outer weld cap (preventing hard spot cracking)	API 5L 10.2.4.8	W
9	Residual Stress Slitting Ring Test	Mandatory 1 test per heat	150 mm wide ring opposite weld	Project Specification	Slitting ring opening: Internal hoop stress <= 0.10 x SMYS (Maximum 55.5 MPa)	API 5L / BOTAS Spec	H/W
10	Mill Hydrostatic Pressure Test	Each pipe (100% all 56" heavy-wall pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 100% SMYS (190.5 bar), Tutma Süresi: MINIMUM 20 SECONDS (+0 / -2 bar calibration)	API 5L 10.2.6.2	H/W
11	Weld Seam 100% Multi-Probe Automated UT	100% full length of SAW weld seam	Full weld seam & HAZ	ISO 10893-11 Level U2	100% multi-channel phased array / pulse-echo AUT + TOFD for mid-wall defect sizing and root fusion	API 5L Ek E	H/W
12	Offline Radiographic Testing (RT)	Pipe ends (200 mm) & repair locations	Weld seam ends and weld repairs	ISO 10893-6 Class B	100% digital radiography or X-ray film Class B; no cracks, lack of fusion, or slag exceeding Class B limits	API 5L Ek E	H/W
13	Full Body Plate & Pipe Ends Laminar AUT	100% full plate area + 100% pipe ends	Plate body & 100 mm pipe ends	ISO 10893-9 / ISO 10893-8	Plate body: 100% scanned (ISO 10893-9 Class E1); Pipe ends: 100 mm circumferential band (no defect > 4 mm)	API 5L Ek E.8	W
14	Automated Laser Geometry & Ovality Station	100% of all pipes (continuous laser)	Pipe body, ends, and weld crown	Optical Laser Profiler	Body OD: +-0.35%; End Out-of-Roundness (Ovality) <= 2.5 mm; Straightness <= 0.10% L; Radial offset <= 1.5 mm	API 5L Tablo 10/16	W
15	Wall Thickness Verification (Heavy-Wall)	100% of all pipes (automated UT scanning)	Pipe body & ends	API 5L 10.2.8.2	Nominal t = 25.40 mm; Tolerance: -5.0% / +10.0% (Acceptance: 24.13 mm to 27.94 mm)	API 5L Tablo 11	W
16	Weighing & Mass Tolerance	100% of all pipes (certified load cells)	Pipe scale	API 5L 9.11.2	Single pipe mass tolerance: -3.5% / +10.0% of theoretical calculated mass (876.3 kg/m)	API 5L 9.11.2	W
17	Visual Surface Inspection & Weld Seam Cap	100% internal and external surface	Full inside & outside pipe	API 5L 10.2.7	100% visual inspection; weld crown height <= 3.0 mm; zero arc burns, scabs, gouges, or slivers	API 5L 9.12	W
18	Residual Magnetism & Final Release	100% of all pipe ends before loadout	Pipe ends (4 quadrature points)	Gaussmeter / Hall effect probe	Residual magnetism: Maximum average <= 2.0 mT (20 Gauss), individual <= 2.5 mT; EN 10204 3.2 cert	API 5L 10.2.8.5	H/W